Keywords: agentic video • world-state memory • cross-shot consistency • video generation

Any Time, Any Space

A World-State-Centric Framework for Agentic Video Generation Maintains Two Persistent Memories—Scene Geometry and Indexed Dynamic Events—to Ensure Full Spatio-Temporal Coherence Across Nonconsecutive Shots

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arXiv:2610.20571 • October 2026

Long-form narrative video generation requires that any shot—requested at any time, depicting any space—remain spatially and temporally consistent with every other shot in the production. ATAS (*Any Time, Any Space*) achieves this by maintaining a *scene memory* of registered static geometry and an *event memory* of dynamic observations indexed by scene, event, and time. Before generating each shot the system retrieves the relevant evidence; after generation a review gate accepts or rejects the clip before registering it into memory. Across four video generators ATAS improves every reported cross-shot consistency metric and tops all six human-rating criteria.

AT A GLANCE

Problem: Long-form agentic video suffers from location drift and temporal contradictions when shots are generated nonconsecutively; existing methods lack a shared spatial and temporal state.

Why it matters: Agentic video is an emerging AI application where coherence failures break narrative trust; no prior approach solves multi-shot spatio-temporal consistency.

Method: Scene memory stores registered geometry per location; event memory stores dynamic observations indexed by scene, event, and time; a review gate decouples film delivery from memory admission.

Result: Highest aggregate benchmark score per backbone; tops all six human-rating criteria across four video generators.

Evidence: Four video generators; spatial revisit, event recapture, and temporal consistency benchmarks; human evaluation on six criteria.

The Big Picture

Narrative video pipelines built from large generative models suffer from *scene drift*: ask the same model to re-enter a room from a new camera angle three shots later, and it will produce a room that contradicts the layout established earlier. The same problem afflicts event recapture—a flashback to an event the viewer already witnessed will not match the original.

These failures arise because every shot is generated independently. The model conditions on a text prompt and, optionally, the most recent frame, but neither source encodes the durable spatial and temporal facts the narrative has established. ATAS resolves this by replacing the stream of frames with a *world-state representation*: two structured memories that persist across the entire production, regardless of generation order.

Why Existing Approaches Fall Short

Prompt conditioning passes shot descriptions between generation calls. It provides no spatial grounding; the model must re-invent geometry from text at each step, and two independent descriptions of the same room will produce inconsistent geometry.

Recent-frame conditioning threads the previous clip's last frame into the current generation. This provides visual continuity for adjacent shots, but cannot support true out-of-order revisits or event recall across large temporal distances in the screenplay.

3D-guided methods impose rigid scene constraints extracted from monocular depth or NeRF reconstruction. These are difficult to update as the narrative introduces new objects, character motion, and destructive events; they also require per-production reconstruction passes that scale poorly.

Agent planning without state improves shot ordering and prompt engineering but leaves individual generation calls unmodified; without a persistent world model, cross-shot coherence is not guaranteed.

Core Mechanism: ATAS

ATAS structures the agentic loop around two memories:

Scene Memory stores a geometric descriptor $G(s_i)$ for each named scene s_i. When a location is first visited, the generated clip is reconstructed and the geometry registered. All subsequent shots at s_i retrieve $G(s_i)$ and condition on it, ensuring consistent spatial layout regardless of camera angle or narrative time.

Event Memory stores a clip descriptor $D(s_i, e_j, t)$ for each combination of scene s_i, event e_j, and time t. When an action or state change is generated, the resulting clip is reconstructed and registered at the appropriate indices. A later shot requesting a replay retrieves the stored entry rather than re-generating from a text description.

Review Gate: Before registering any clip, ATAS applies quality and state-consistency checks. A rejected clip is regenerated; film delivery is independent of memory admis-

sion, so the output stream never blocks and hallucinated geometry is never registered.

Technical Formulation

Let $\mathcal{S} = \{s_1, \ldots, s_N\}$ be named scenes and $\mathcal{E} = \{e_1, \ldots, e_M\}$ be narrative events. Scene memory maps each scene to its registered geometry:

$$G(s_i) = \text{Reconstruct}\Big(v^{(1)}(s_i)\Big)$$

where $v^{(1)}(s_i)$ is the first accepted clip at scene s_i.

Event memory maps a triple to a dynamic descriptor:

$$D : \mathcal{S} \times \mathcal{E} \times \mathbb{R}_{\geq 0} \to \mathcal{D}$$

For a shot request $r = (s_i, e_j, t, \text{cam})$, the conditioning evidence is the concatenation of the two retrieved descriptors:

$$C(r) = G(s_i) \oplus D(s_i, e_j, t)$$

and the generator produces:

$$v_r = \text{VideoGen}(r,\ C(r))$$

After the review gate accepts v_r, the event memory is updated:

$$D(s_i, e_j, t) \ \leftarrow\ \text{Reconstruct}(v_r)$$

If the review gate rejects v_r, the clip is regenerated; the memory state is unchanged, preventing error propagation.

Learning or Inference Procedure

1. Parse the screenplay into shot requests $r = (s_i, e_j, t, \text{cam})$.
2. For each request, query scene memory for $G(s_i)$ and event memory for $D(s_i, e_j, t)$; retrieve empty descriptors for unseen entries.
3. Condition the video generator on $C(r)$ and generate clip v_r.
4. Apply the review gate; if accepted, update $D(s_i, e_j, t)$ and $G(s_i)$ (on first visit); deliver v_r to the output stream.
5. If rejected, return to step 3 with a refreshed generation seed.
6. Repeat until all shots are delivered.

No additional model training is required. ATAS is a retrieval-and-conditioning framework that wraps any video generation backbone.

What the Guarantee Says

ATAS does not make formal guarantees on per-shot quality, which depends on the underlying generator. Its empirical claims are:

- Every reported cross-shot consistency metric improves across all four tested video generators when ATAS is added.
- ATAS achieves the highest aggregate benchmark score per backbone on the spatial-revisit and event-recapture tasks.
- Human raters score ATAS highest on all six criteria (narrative coherence, spatial accuracy, temporal consistency, visual quality, event fidelity, audio-visual alignment) across all tested generators.

Experimental Findings

- **Backbones:** Four video generators covering diffusion-based and autoregressive architectures.
- **Spatial revisit:** ATAS outperforms compared agent-based and 3D-guided methods on the target-image spatial revisit task.
- **Event recapture:** ATAS leads on three event-recapture criteria; prior methods degrade to text-driven re-generation.
- **Temporal consistency:** All temporal-consistency metrics improve under ATAS for every backbone.
- **Human evaluation:** ATAS tops all six human-rating criteria across all tested generators; previous methods trail on at least two.
- **Out-of-order generation:** ATAS maintains consistency even when shots are generated out of screenplay order.

Ablations and Interpretation

Without scene memory: Spatial layout drifts across revisit shots; objects disappear or change position; the spatial-revisit score drops sharply.

Without event memory: Event recapture degrades to text-driven re-generation; temporal positions shift; event-fidelity scores drop.

Without review gate: Hallucinated geometry is registered and propagates to subsequent shots; errors compound as the production grows.

Memory granularity: Finer time indexing in event memory yields better event recapture; coarser indexing reduces memory overhead at the cost of temporal precision.

Why retrieval beats prompting: Geometry extracted from accepted clips encodes exact object positions that a text description cannot; conditioning on this evidence directly constrains the generator's spatial output distribution, while text descriptions leave spatial degrees of freedom underdetermined.

Reference

Weijie Wang, Haomin Zhang, Bohan Zhuang, et al. **Any Time, Any Space: A World-State-Centric Framework for Agentic Video Generation**. arXiv:2610.20571, October 2026.
https://arxiv.org/abs/2610.20571